

FUEL MANAGEMENT SYSTEM

AN INTERNSHIP REPORT

Submitted by

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190950131063

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

In

**Computer Science and Engineering
ITM Universe, Vadodara**



Gujarat Technological University, Ahmedabad

[May, 2023]



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CERTIFICATE

This is to certify that the project report submitted along with the project entitled **FUEL MANAGEMENT SYSTEM** has been carried out by **Patel Arth KalpeshKumar** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering, 8th Semester of Gujarat Technological University, Ahmadabad during the academic year 2022-23.

Prof. Palak Vashi
Internal Guide

Prof. Gaurav Kulkarni
Head of the Department

COMPLETION CERTIFICATE



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28th April 2023

Mr. Arth Patel
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Subject: Internship Completion certificate

TO WHOM SO EVER IT MAY CONCERN

This is to certify that **Arth Patel**, final year student of **ITM University, CSE** has successfully completed his Internship with **Baroda Informatics Pvt. Ltd.** for a period of Three Months from **1st Feb'23 to 30th Apr'23**.

During the internship he was assigned the project on "**Fuel Management System-8th version**" modules covering the Admin (Active & Inactive Groups), Contract management, Cargo Bank, Counterparty Master and Incident Tracker report in FMS – NG using JavaScript, CSS and HTML.

During the period of his internship program with us, we have found him to be prompt, hardworking and willingness to learn.

We take this opportunity to wish him a very bright and successful career ahead.

For **Baroda Informatics Pvt. Ltd.**

Sutanu Chowdhury
Lead - HR



ITM UNIVERSE, VADODARA

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DECLARATION

I hereby declare that the Internship / Project report submitted along with the Internship/ Project entitled **Fuel Management System** submitted in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me / us at **Baroda Informatics Pvt. Ltd** under the supervision of **Mrs. Renu Kumari** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Sign of Student

1

Patel Arth KalpeshKumar

ACKNOWLEDGEMENT

I am giving hearty thanks to **Baroda Informatics Pvt. Ltd** for providing me this opportunity for taking this kind of Internship Program which would be an asset for me for my future carrier and I am also very thankful to all who have directly or indirectly supported me in completion of this Internship Program. Proper organization of concept and analysis of the system is due to keen interest and helping hand of my faculties and colleagues.

I am taking this opportunity to express my gratitude and to thank all they helped me in this Internship Program. The whole process of internship was able to bring out much more than what was expected by me, out of me. “Learning is a continuous process” and at this moment, I hardly have enough words to express my gratitude toward those who were constantly involved with me during my Internship Program period.

The guide performs their role of path shower. Under the shade of proper guidance, a person can shape his untouched talent. Thus, my special obligations remain due towards **Mr. Hiren Ahir , Mrs. Renu Kumari & Mr. Saibal Bhowmik** for continuously guiding me at the company and answering all my doubts with patience. I would also like to thank you my Internal guide **Prof. Palak Vashi** for helping us through our internship by giving us the necessary suggestions and advices along with their valuable co-ordination in completing this internship.

Finally, I am thankful to my friends who were always there to support and help me whenever needed. I once again thank the staff members of **Baroda Informatics Pvt. Ltd** for their valuable support in completion of the Internship.

Thank You

Arth Patel (190950131063)

Internal Guide: prof. Palak Vashi

HOD: Prof. Gaurav Kulkarni

ABSTRACT

Project Title : Fuel Management System

Key Points:

- System provides efficient fuel management facility to different entities such as Customer, Transporter, Supplier and Vendor.
- Fuel Management System deals with Entities initiating Fuel requirement from Energy Suppliers by entering into contract.
- Customers are first created as an Entity in Fuel Management System and then deal is signed between Customer and Supplier for fuel supply as per Customer's requirement.

Fuel Management System is a fuel management software for supply and store of fuel where users initiate their requirement to energy suppliers by entering into a buy/sell deal. Fuel management system consists of different modules containing Forms & Reports.

All Modules, Menu Items and the corresponding Forms / reports provide user-friendly interface which is consistent throughout the application. As applicable, the validation checks are also mentioned to enable end users to avoid operational mistakes. End users operate FMS to carry out their own function specific tasks.

Technologies:

- Apache Tomcat 8.0,
- J2EE,
- Oracle 12c (Database)

Project Assignment:

- Enhancement & Technical Support.

LIST OF FIGURES

Sr. No	Figures	Page No.
1.	3.5 (a) J2EE	10
2.	3.5 (b) Oracle 12c	11
3.	4.5 (a) Flow Chart of Login.....	20
4.	4.8 (a) Agile Software Development.....	24
5.	5.2.1 (a) State Transition Diagram.....	26
6.	5.2.1 (b) ER Diagram.....	27
7.	5.2.2 (a) Login Page.....	28
8.	5.2.2 (b) Home Page	28
9.	5.2.2 (c) Add/Modify Group	29
10.	5.2.2 (d) Add/Modify User	29
11.	5.2.2 (e) Group Allocation Delegation	30
12.	5.2.2 (f) User Details	30
13.	5.2.2 (g) Access Right Administration	31
14.	5.2.2 (h) Cargo Bank	31
15.	5.2.2 (i) Supply Notice	32
16.	5.2.2 (j) Seller's Nomination To Customer	32
17.	5.2.2 (k) Fortnightly Buyer Nomination	33
18.	5.2.2 (l) Fortnightly Buyer Nomination Details	33
19.	5.2.2 (m) Lock User Page	34
20.	5.2.3 (a) FMS7_SN_TRANSPORTER_DTL	34
21.	5.2.3 (b) HR_EMP_MST	35
22.	5.2.3 (c) SEC_EMP_GRP_DTL	35
23.	5.2.3 (d) FMS7_TRANSPORTER_MST	36
24.	5.2.3 (e) SEC_ACCESS_GROUP_MST	36
25.	5.2.3 (f) FMS7_FN_BUYER_NOM_DTL	37
26.	5.2.3 (g) SEC_FORM_MST	37
27.	5.2.3 (h) SEC_EMP_PASSWORDS.....	38
28.	5.2.3 (i) FMS9_COUNTERPARTY_MST	38
29.	6.2 (a) cross platform	40

ABBREVIATIONS

- **BIPL** : Baroda Informatics Pvt. Ltd
- **FMS** : Fuel Management System
- **SRS** : System Requirement System
- **BRS** : Business Requirement System
- **FAT** : Functional Acceptance Test
- **UAT** : User Acceptance Test
- **J2EE** : JAVA 2 Platform, Enterprise Edition
- **IDE**: Integrated Development Environment

TABLE OF CONTENTS

Acknowledgement	i
Abstract	ii
List of Figures	iii
List of Abbreviations	iv
Table of Contents	v
1. Overview of Company	1
1.1. History of Company	1
1.2. Scope of Work	1
1.3. More about Company's Information	1
1.4. Company Vision	2
1.5. Company Objective	2
2. Overview of different plant/unit/department/shop of the organization and Layout of the production/process being carried out in company	4
2.1. It includes the details about the work being carried out in each department.....	4
2.2. List the technical specifications of major equipment used in each department	4
2.3. The process of developing an end product involves the following sequence of operations.	4
2.4. Explain in details about each stage of production.	5
3. Introduction to Project / Internship	7
3.1. Summary of Project / Internship	7
3.2. Purpose	7
3.3. Aim and Objective of Project / Internship	8
3.4. Scope	9
3.5. Technology and Literature Review	9
3.6. Internship Project Planning	12
3.6.1. Internship Project Development Approach and Justification	12
3.6.2. Roles and Responsibilities.....	13
3.6.3. Risk Analysis	13
3.6.4. Group Dependencies	13
4. System Analysis	14
4.1. Study of current System	14
4.2. Problem and Weaknesses of Current System	15
4.3. Requirements of New System	15
4.4. System Feasibility	16
4.4.1. Does the system contribute to the overall objectives of the organization?	16

4.4.2. Can the system be implemented using the current technology and within the given cost and schedule constraints?	17
4.4.3. Can the system be integrated with other systems which are already in place?	18
4.5. Process in New System / Proposed System	18
4.6. Features of New System	20
4.7. List of Main Modules	21
4.8. Selection of Technology	22
5. System Design	25
5.1. System Design and Methodology	25
5.2. Input/ Output and Interface Design	26
5.2.1. State Transition Diagram.....	26
5.2.2. Sample of Interface	28
5.2.3. Database Tables	34
6. Implementation	39
6.1. Implementation of Platform / Environment	39
6.2. Process / Technology / Module Specifications	40
6.3. Results / Outcomes	40
6.4. Result Analysis	40
7. Testing	41
7.1. Testing Plan / Strategy	41
7.2. Test Result and Analysis	42
7.2.1 Test Cases	42
8. Conclusion and Discussion	43
8.1. Conclusion / Overall Analysis	43
8.2. Photographs and date of surprise visit by institute mentor	43
8.3. Dates of Continuous Evaluation	43
8.4. Problem Encounter and Possible Solution	44
8.5. Summer of Internship Project	44
8.6. Limitation and Future Enhancement	45
9. References	47

1. Overview of Company

Baroda Informatics Private Limited are pioneer in providing solution to the national Gas Users. Over the years, we have worked closely with national Natural-Gas Users and developed a standardized web-enabled solution which is platform independent, modular, scalable and user friendly. The Team of Engineers has designed, developed and successfully implemented this solution, thus turning the dream concept of Fuel Management into a reality solution.

1.1. History of Company

BIPL was established in 2007 primarily as a Product Company offering Software Solutions and ITeS in certain niche markets. Within a short span of five years, the company has succeeded in making its presence felt in these market segments. Within this time, the company has attained a significant increase in its turnover and grown to a current team size of 25 professionals coming from various disciplines. The company has also established its presence along with the necessary infrastructure, at 3 locations in India. Baroda, Kolkata & Mysore. BIPL is aggressively focused on Software / Applications Development together with Infrastructure Management, with major projects under way.

1.2 Scope of Work

Baroda Informatics Pvt. Ltd is Drawing from the rich experience of its promoters, BIPL follows a well laid down methodology for its activities. BIPL has a definite roadmap for achieving, Quality Certification both for its development and Service offerings. BIPL have good Chance to work in full Stack Web Development using Core java.

1.3 More about Company's Information

The mission of BIPL is to conceptualize, design and develop IT solutions specific to customer requirement. It also provides support &

service in field of Hardware, Software, Networking and Facility Management throughout India. The operations also include help desk support and in house Training. BIPL is dedicated to the concept of continuous improvement. We initiate research and development of new innovative ideas and products that can assist Indian community as a whole. In addition, BIPL's Training Delivery Division is dedicated to the development and support of Learning System for Engineers, both in the field of development of software.

- **Achievements of Company**
 - **2007:** When the Company Started
 - **15+** Global Customers
 - **25+** IT Professionals
 - **3** Cities with active customers
 - **10+** Completed Projects

1.4. Company Vision

The vision of BIPL is to conceptualize, design and develop IT solutions specific to customer requirement. It also provides support & service in field of Hardware, Software, Networking and Facility Management throughout India. The operations also include help desk support and in house Training. BIPL is dedicated to the concept of continuous improvement. We initiate research and development of new innovative ideas and products that can assist Indian community as a whole.

1.5. Company Objective

- Adaptability to emerging trends and technology in the market
- Understanding of existing modules or incorporation of new module

- Reduction time in Collecting/Receiving various data
- Reduction in instances of manual error
- Availability and reliability of Information Processing/Collection
- Improvement in equipment and man power utilization from time to time
- Increase in Productivity
- Improvement in business performance
- Conducting of Audits and reporting more easily
- Availability of current and historical information at the touch of a button
- Traceability of On-line data
- Improvement in efficiency by streamlining existing and removing bottlenecks
- Secured and controlled access by Administrator only
- Access log of all activities .

2. Overview of different plant/unit/department/shop of the organization and Layout of the production/process being carried out in company.

2.1. It includes the details about the work being carried out in each department.

Baroda Informatics Pvt. Ltd is organized into three main departments: Support and Services, Customized Software Development, and Testing. The Customized Software Development department handles all technical aspects of software development and specializes in creating fully customizable software tailored to meet clients' specific needs. The Support and Services department provides comprehensive technical and non-technical support, including Annual Maintenance Contracts (AMCs). The Testing department collaborates with the developers to ensure software quality, and provides detailed feedback to the development team.

2.2. List the technical specifications of major equipment used in each department.

The Software Development department utilizes various technical specifications and essential equipment, such as a central server based entirely on Linux, as well as Oracle and MySQL databases.

2.3. The process of developing an end product involves the following sequence of operations:

- Gathering requirements from the client
- Conducting a Business Requirement Study (BRS)
- Developing the software
- Testing the software
- Deploying the software for the end user

2.4. Explain in details about each stage of production:

2.4.1. Gathering requirements from the client:

- This is the first stage in software production where the software development team meets with the client to identify and gather all the requirements for the project. The development team works closely with the client to fully understand their needs and objectives. During this stage, the team asks questions, takes notes, and prepares a project plan based on the client's requirements.

2.4.2. Conducting a Business Requirement Study (BRS):

- After gathering the client's requirements, the development team conducts a Business Requirement Study (BRS). This stage involves analyzing the gathered information and breaking it down into smaller, more manageable parts. The team identifies any gaps in the requirements, confirms the accuracy of the information received from the client, and verifies the feasibility of the project. This stage helps ensure that the project can be completed on time, within budget, and to the client's satisfaction.

2.4.3. Developing the software:

- The development stage is where the actual coding of the software begins. The development team uses the client's requirements and the BRS as a guide to design and build the software. The team works on developing different components of the software and ensures that they function correctly and integrate with other parts of the software.

2.4.4. Testing the software:

- Once the software is developed, the next stage is testing. The development team tests the software to verify that it meets the client's requirements and works as intended. This stage includes identifying and fixing any bugs or errors, optimizing the software's performance, and ensuring that the software is stable and reliable.

2.4.5. Deploying the software for the end user:

- The final stage is deploying the software for the end user. This stage involves installing and configuring the software on the client's system and ensuring that it works correctly. The development team provides the client with all the necessary training, support, and documentation to ensure that the client can use the software effectively. Once the client has approved the software, it is officially launched and made available to the end-user.

3. Introduction to Internship Project

The internship program is designed to provide students engaged in a field experience with an opportunity to share their insights, to explore the links between students' academic preparation and their field work, and to assist participants in developing and carrying out the major research project which will serve to culminate their internship experience.

Internships are individualized and tailored to the needs and interests of each student in the program. As part of the internship experience, students are expected to take an active role in finding an appropriate internship for themselves. Many students pursue their own contacts; however, information is available on Locating Internship Sites.

3.1. Summary of Internship Project

A Fuel Management System is a fuel management software for supply and store of fuel where users initiate their requirement to energy suppliers by entering into a buy/sell deal. Fuel management system consists of different modules containing Forms & Reports.

3.2. Purpose

A Fuel Management System is a fuel management software for supply and store of fuel where users initiate their requirement to energy suppliers by entering into a buy/sell deal. Fuel management system consists of different modules containing Forms & Reports. System provides efficient fuel management facility to different entities such as Customer, Transporter, Supplier and Vendor. Fuel Management System deals with Entities initiating Fuel requirement from Energy Suppliers by entering into contract. Customers are first created as an Entity in Fuel Management System and then deal is signed between Customer and Supplier for fuel supply as per Customer's requirement.

3.3. Aim and Objective of Internship / Project

- **Performance Test**

One of the core reasons and purposes of an internship is to expose you to a specific job, profession, or industry. While you may have an idea of what a job is, it is only when you do it that you will know if this is what you think, whether you have the education and skills to do it, and if you like it. For example, you might think that marketing is a creative process that involves creating a slogan and running a Campaign. An internship with an advertising agency will help you find out that advertising includes consumer demographics, focus groups, knowledge of customer pricing and sales strategies, and media research and buying.

- **Network Contacts**

Another advantage of being an internship is the building of business contacts. These people can help you find a job later, act as a referral, or help you with a project after you've been hired elsewhere. Meet the people who have the jobs you want one day and ask if you can take them out to lunch. Ask them how they initiated in their career, how they got to where they are, and if they have suggestions for improving your skills.

- **Increase Your Confidence**

An internship will, of course, help you find out more about the work environment but also about yourself. You have a clear idea of your strengths, weaknesses, likes, and dislikes. Knowing that you have hands-on experience will make you more confident in job hunting and interviews.

- **Possible Work**

If you enjoy internship, you have the opportunity to differentiate yourself and end up working for a company. Employers find it way smoother and easier to hire someone they know because they can sense that person's work ethic, skills, creativity, and ability to work with other people and generally fit in with the company. Do more than necessary, study office policies, and make friends

with important people to increase your chances of starting a business.

3.4. Scope of Internship Project

An Internship Provides Real Life Experience and Exposure. The Opportunity to Learn More about Yourself. Get Connected and Develop Your Professional Network. Prevent CV From Going to The Trash. Transition into a Full-Time Position. And, A Fuel Management System is a fuel management software for supply and store of fuel where users initiate their requirement to energy suppliers by entering into a buy/sell deal. Fuel management system consists of different modules containing Forms & Reports.

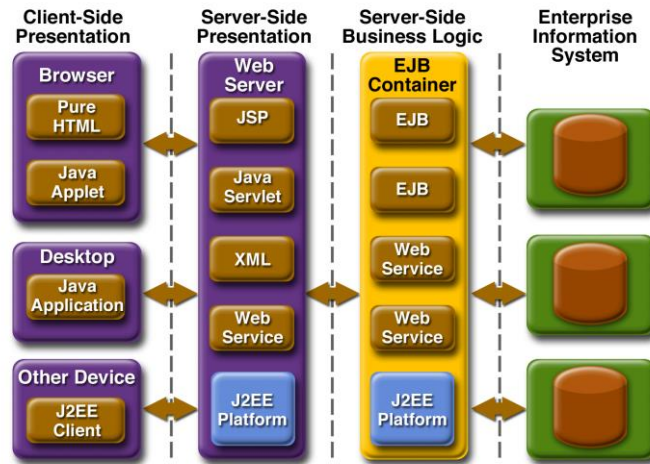
System provides efficient fuel management facility to different entities such as Customer, Transporter, Supplier and Vendor. Fuel Management System deals with Entities initiating Fuel requirement from Energy Suppliers by entering into contract. Customers are first created as an Entity in Fuel Management System and then deal is signed between Customer and Supplier for fuel supply as per Customer's requirement.

3.5. Technology and Literature Review

For Development of this web Application, we use Core Java , HTML , CSS , JS & Bootstrap.

Tools been used in development are,

- 3.5.1.** Eclipse IDE
- 3.5.2.** Apache Tomcat server
- 3.5.3.** J2EE
- 3.5.4.** Oracle 12c (Database)

Technology used,**J2EE :**

[Fig 3.5(a) J2EE]

The Enterprise Java BluePrints for the J2EE platform describe the J2EE application model and best practices for using the J2EE platform. Building on the J2SE platform, the J2EE application model provides a simplified approach to developing highly scalable and highly available internet or intranet based applications. The J2EE platform provides choices for graphical user interfaces across a company's intranet or on the World Wide Web. Clients can run on desktops, laptops, PDAs, cell phones, and other devices. Pure client-side user interfaces can use standard HTML and Java applets. Support for simple HTML means quicker prototypes, and support for a broader range of clients. Additionally, the J2EE platform supports automatic download of the Java Plug-in to add applet support where it's lacking. The J2EE platform also supports stand-alone Java application clients.

Oracle 12c (Database) :

[Fig 3.5(b) Oracle 12c]

Oracle released Oracle Database 12c into general availability July 1, 2013. According to Oracle, this is "the first database designed for the cloud." Oracle Database 12c also introduces 500 new features to the database, most notably pluggable databases and multitenant architecture.

Oracle Database 12c is a relational database management system (RDBMS) designed for both on-premises and cloud uses and deployable on a choice of clustered or single servers. It provides comprehensive features for managing data in transaction processing, business intelligence and content management applications.

An added benefit of performing the migration in Oracle Database 12c is that the new extended VARCHAR2 type limit of 32767 bytes guarantees any VARCHAR2 column with the pre-12.1 limit of 4000 bytes or less can just be lengthened to accommodate longer values resulting from data expanding in conversion to Unicode.

3.6. Internship Project Planning

- Project planning is part of project management, which relates to the use of schedules such as Gantt charts to plan and subsequently report progress within the project environment.
- Initially, the project scope is defined and the appropriate methods for completing the project are determined. Following this step, the durations for the various tasks necessary to complete the work are listed and grouped into a work breakdown structure.
- Project planning is often used to organize different areas of a project, including project plans, workloads and the management of teams and individuals.

3.6.1. Internship Project Development Approach and

Justification Incremental Model design

The incremental model is a development approach that involves dividing a project into smaller, more manageable parts or increments. Each increment is developed and tested separately, and once it is complete, it is integrated with the previous increments to form a complete system. This approach allows for more flexibility and easier management of changes, as well as faster delivery of working software. For an internship project, the incremental model can be an effective approach for several reasons:

- It allows for continuous feedback and improvement: Since each increment is developed and tested separately, feedback can be gathered from stakeholders and incorporated into the next increment. This ensures that the final product meets the needs and expectations of all stakeholders.
- It reduces the risk of failure: By developing and testing each increment separately, any issues or errors can be identified and addressed before they become major problems. This reduces the risk of the project failing due to unforeseen issues.
- It allows for faster delivery: Because each increment is developed and tested separately, it can be delivered to stakeholders as soon as it is complete. This means that stakeholders can start using and benefiting from the software earlier in the development process.
- It provides a clear roadmap for development: By dividing the project into smaller increments, the development team can focus on developing and

testing each increment separately, which provides a clear roadmap for development and helps to ensure that the project stays on track.

3.6.2. Roles and Responsibility

There is role of the web developer to first of all analysis, designing, coding, testing and documentation. The main responsibility of the system is to satisfy the requirements and provide the all solution their query. The main responsibility is that we had to match all the deadlines.

3.6.3. Risk Analysis

Project Risk Management involves conducting risk management planning, engaging in risk identification, completing risk analysis, creating a risk response action plan, and monitoring and controlling risk on a project. Project Risk Management is a continuous process to be engaged in throughout the entire project. A key point to remember is that risk is not always bad. There are opportunities and there are threats. The opportunities are the good risks. The treats are the bad risks. Risk Identification is a systematic attempt to specify threats to the project plan. By identifying the known and predictable risks, the project manager takes a first step towards avoiding them when possible and controlling them when necessary one method of identifying risks is to create a risk item checklist. Regardless of the prevention techniques employed, possible threats that could arise inside or outside the organization need to be assessed. Although the exact nature of potential disasters or their resulting consequences are difficult to determine.

3.6.4. Group Dependencies

The project was handled by 9 people in a group so we are worked in the group on project our senior developer working with database. If I was stuck anywhere, I used to approach directly to my mentor.

4. System Analysis

4.1. Study of current System

- For Fuel Management System the context of customized software equipment, a study of the current system would involve a thorough examination of the software, including its features, functionality, user interface, and performance. This analysis is typically conducted by a software engineer or a team of engineers who have expertise in the programming language and development tools used in the software.
- The study would begin with a review of the software's documentation, including the user manual and technical specifications. The engineers would also review the source code of the software to understand how it is structured and how the various components interact with each other.
- Once the engineers of the company get a good understanding of the software, they would then assess its strengths and weaknesses. This would involve identifying any bugs or glitches in the software, as well as any performance issues or user interface problems. They would also evaluate the software's ability to meet the needs of its users, based on user feedback and any other available data.
- Based on their analysis, the engineers would then recommend modifications or improvements to the software. These changes might include bug fixes, performance optimizations, or updates to the user interface. The engineers would then work with the software development team to implement these changes and test them thoroughly before releasing the updated software to users.
- It is important to note that customized software equipment is often developed specifically for a particular user or organization. This means that the software may be modified or updated on a regular basis to meet the evolving needs of its users. As such, the study of the current system is an ongoing process that may be repeated periodically to ensure that the

software continues to meet the needs of its users over time.

4.2. Problem and Weakness of Current System

When evaluating the problem and weaknesses of a current system, one would typically start by examining the system's purpose, design, and functionality. This would involve looking at the system's features, how it operates, and how it interacts with other systems or processes. Through this evaluation, one can identify areas where the system falls short, such as:

- 1. Technical issues:** These include bugs, glitches, and errors that hinder the system's performance or cause it to crash or freeze.
- 2. Inefficient processes:** The system may have inefficient processes or workflows that lead to delays or errors in completing tasks.
- 3. User experience:** The system may have a poor user interface or lack essential features that make it challenging or frustrating for users to navigate and use effectively.
- 4. Data accuracy:** The system may have data inconsistencies or inaccuracies, leading to incorrect information and reports.
- 5. Scalability:** The system may not be scalable, meaning it cannot handle a large volume of users or data.

4.3. Requirements of New System

- As technology advances and businesses evolve, there is a constant need to adapt and improve existing systems to meet changing requirements. Adding new modules and functionalities to a system can greatly enhance its capabilities and effectiveness, allowing it to better serve the needs of the users.
- In the case of a system that requires new modules such as buyer nomination, fetching data fortnightly, and auto mail generation, these additions could have a significant impact on the overall functionality and efficiency of the system. For example, the ability to automatically generate emails can save users a significant amount of time and effort, freeing them up to focus on more important tasks.
- Similarly, the ability to fetch data fortnightly can provide users with more up-

to-date information, enabling them to make more informed decisions. The addition of a buyer nomination module can help streamline the buying process, making it easier for both buyers and sellers to complete transactions.

- Ultimately, the need for a new system with added modules and functionalities arises from the desire to improve efficiency, accuracy, and overall effectiveness. By continuously updating and improving systems, businesses can better meet the needs of their users and remain competitive in an ever-changing market.

4.4. System Feasibility

4.4.1. Does the system contribute to the overall objectives of the organization?

Yes, the system contributes to the overall objectives of the organization as it is a business model, the entire organization depends on it.

In the context of a business model, a system is often developed to support the organization's core functions, processes, and goals. It can help the organization to:

1. Increase efficiency: A well-designed system can help streamline processes, reduce errors, and improve productivity, which can lead to increased efficiency.

2. Enhance customer experience: A system that is customer-centric can help improve customer satisfaction by providing a more personalized and efficient service.

3. Improve decision-making: A system that provides accurate and real-time data can help management make informed decisions that drive the organization towards its objectives.

4. Support growth: A system can help support the organization's growth by providing scalability and flexibility, allowing it to adapt to changing market conditions and expand its operations.

5. Reduce costs: A system that automates processes and reduces the need for manual labor can help reduce costs and improve profitability.

Overall, a system can help an organization achieve its goals by providing a structured and organized approach to its operations. It can help align the organization's resources, processes, and activities towards achieving its objectives, which in turn can contribute to its success and growth. Therefore, it is essential to ensure that the system is designed and developed in a way that supports the organization's overall objectives and is continuously evaluated and optimized to meet changing business needs.

4.4.2. Can the system be implemented using the current technology and within the given cost and schedule constraints?

Yes, it is possible to implement the system using the current technology if the technology being used is platform-independent and the system's functionality is primarily dependent on the front-end.

Platform-independent technology would allow the system to be run on multiple platforms and devices, without requiring significant modifications to the underlying code. This can help to ensure that the system is accessible to the widest possible audience and can be easily scaled up or down as needed.

Additionally, if the system's functionality is primarily dependent on the front-end, it means that the core logic and business rules can be abstracted away from the implementation details. This can make it easier to modify and enhance the system over time, without requiring major changes to the underlying architecture or backend code.

While the quality of work is certainly an important concern for any business, it's important to note that time and cost are also critical factors that need to be taken into consideration when developing a system. The system's effectiveness will be measured not just by the quality of its output, but also by how efficiently it can be developed and maintained over time, and how much it costs to do so.

Choosing not to use a framework can have implications for both time and cost. Frameworks are designed to provide a set of pre-built tools and

components that can speed up the development process, reducing the amount of time and effort required to create a system. Without a framework, developers may need to create these components from scratch, which can be a time-consuming process.

Furthermore, frameworks can help ensure that code is maintainable and scalable over time, reducing the likelihood of bugs and making it easier to add new features and functionality as the system evolves. Without a framework, it may be more difficult to maintain the codebase, leading to higher costs over time.

Ultimately, the choice of whether to use a framework should be driven by the specific requirements of the business and the system being developed. While frameworks can be useful tools for speeding up development and improving code maintainability, they are not always necessary, and some systems may be better served by a more customized approach. However, it's important to carefully weigh the tradeoffs between time, cost, and quality when making this decision.

4.4.3. Can the system be integrated with other systems which are already in place?

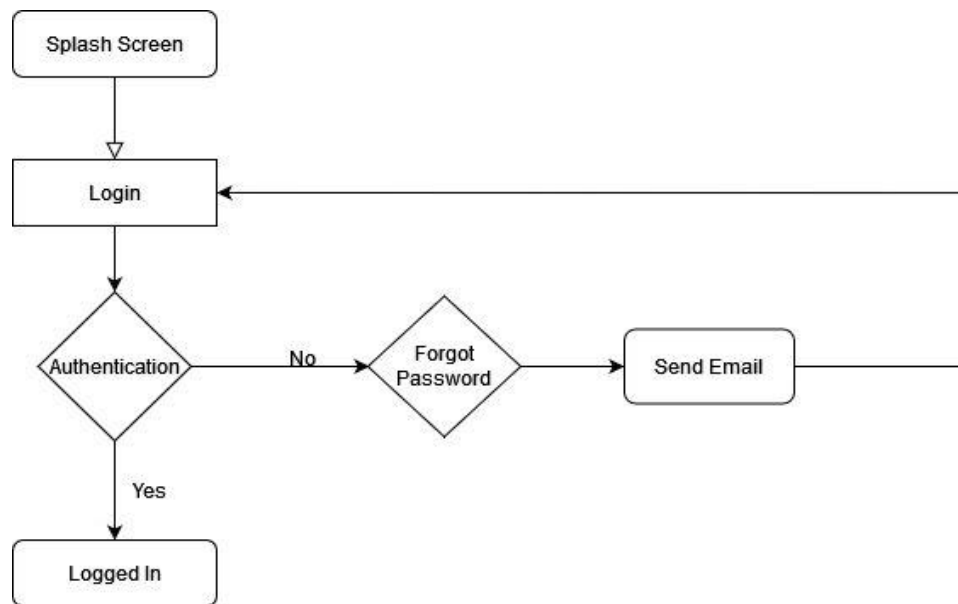
The system can be integrated with other systems which are already in place.

4.5.Process in New System / Proposed System

- Procedure for a new project that involves the system development life cycle:
- Planning Phase: Define the project scope, goals, objectives, budget, and timelines. Determine the resources required for the project and identify the project stakeholders.
- Analysis Phase: Analyze the current system, identify its strengths and weaknesses, and define the requirements for the new system. Develop use cases, functional requirements, non-functional requirements, and system

specifications.

- Design Phase: Develop the design for the new system, including the system architecture, data models, user interface, security, and integration requirements. Define the system components and their interdependencies.
- Implementation Phase: Build and configure the system components according to the design. Develop code, scripts, and workflows. Conduct integration testing to ensure that the system components work together as expected.
- Testing Phase: Conduct various types of testing to ensure that the system meets the requirements and specifications. This includes unit testing, integration testing, system testing, acceptance testing, and performance testing.
- Deployment Phase: Deploy the new system into the production environment. Install the system components and ensure that they work correctly in the live environment. Develop a rollout plan to minimize any impact on the users.
- Maintenance Phase: Maintain the system over its lifecycle. Identify and fix any issues, implement system enhancements, and provide support to the users. Conduct periodic reviews to ensure that the system continues to meet the needs of its users.

Process / Flow chart:

[Fig: 4.5(a) Flow Chart of Login]

4.6. Features of New System

The new system you mentioned appears to be a comprehensive infrastructure for Fuel Management System, which includes the following features:

- 1. Inventory Storing:** This feature involves storing natural gas in large tanks or other storage vessels before it is distributed to customers. The inventory storing system allows for efficient and reliable management of the supply of gas.
- 2. Regasification:** Natural gas is often transported in a liquefied form and needs to be regasified before it can be used. The regasification feature of the system involves the process of converting liquefied natural gas (LNG) back into its gaseous form so that it can be transported through pipelines.
- 3. Pipeline:** The pipeline is an essential part of the system, which allows for the transportation of natural gas from one location to another. The pipeline system ensures that natural gas is delivered safely and efficiently to its destination.
- 4. Transportation:** This feature involves the transportation of gas from the

storage facility to its end-users. It may involve the use of trucks, railcars, or other means of transportation.

5. Billing: The billing feature is essential for managing the financial aspects of the system. It involves the process of generating invoices, calculating costs, and managing customer accounts.

6. Report: The reporting feature involves generating reports and analyzing data related to the performance of the system. It allows for the identification of areas that need improvement and the optimization of the system's overall efficiency.

In summary, the new system includes inventory storing, regasification, pipeline, transportation, billing, and reporting features, all of which are crucial for the efficient management of natural gas.

4.7. List of Main Modules

- Main Modules of the Application are following,

1. Administration: The Administration module is responsible for managing the system's overall configuration, security, and user access control. It allows authorized personnel to add, edit, and delete user accounts, roles, and permissions, as well as manage system settings such as database connection, backup, and restore.

2. Helpdesk: The Helpdesk module is responsible for managing customer support tickets and inquiries. It allows customers to submit tickets or inquiries related to their accounts, invoices, or incidents, and provides a centralized platform for the helpdesk staff to manage, track, and respond to these tickets.

3. Invoice: The Invoice module is responsible for managing the billing and payment processes of the system. It allows the system to generate and send invoices to customers, as well as track and manage payments, credits, and refunds.

4. Incident Portal: The Incident Portal module is responsible for managing and tracking incidents related to the natural gas system. It allows users to report

and track incidents such as gas leaks, pipeline damages, or equipment failures, and provides a centralized platform for the system operators to manage, track, and respond to these incidents.

There may be additional modules or functionalities included in the system, depending on the specific requirements and needs of the customer. These modules may include features such as inventory management, regasification, pipeline monitoring, transportation, reporting, or analytics.

4.8. Selection of Hardware

The selection of tools and hardware configurations should be based on the specific requirements of the customer. The hardware configuration of the server, for example, should be tailored to meet the demands of the system and the workload it will be handling. This may include the number of cores, amount of RAM, storage capacity, and network bandwidth.

Similarly, the choice of software tools used to manage the system should also be based on the specific needs and requirements of the user. For example, different tools may be preferred for managing inventory, regasification, pipeline, transportation, billing, and reporting functions depending on the nature and complexity of the system.

Selection of Software

- Java is a widely used programming language that is known for its platform independence, robustness, and scalability. It is commonly used for building enterprise-level applications, which makes it a suitable choice for the backend of the natural gas management system you described.
- Oracle is a popular database management system that is widely used in enterprise-level applications due to its reliability, security, and scalability. It is often used in conjunction with Java applications, making it a suitable choice for the backend database of the system.

- The front-end of the system is built using HTML, CSS, JavaScript, Bootstrap, and jQuery. HTML is used for creating the structure and content of web pages, while CSS is used for styling and formatting the appearance of the pages. JavaScript is used for creating interactive and dynamic elements on the pages, while bootstrap and jQuery provide pre-built components and functionality to enhance the user interface and user experience.

Overall, the combination of Java, Oracle, HTML, CSS, JavaScript, Bootstrap, and jQuery provides a robust and scalable platform for building the natural gas management system you described.

Approaches and Justification

- If the customer has not specified any approach to follow for the development of the natural gas management system, then the development team can choose the approach that best suits the project's requirements and constraints.
- There are several software development approaches that can be used, each with its own advantages and disadvantages. The approach used by Fuel Management System is Agile model.
- Agile: The Agile approach is an iterative and incremental approach to software development. It is often used when the requirements are not well-defined or may change frequently. This approach involves working in short cycles or sprints, with frequent feedback and collaboration between the development team and the customer.
- The selection of the approach should be based on the specific requirements and constraints of the project, such as budget, timeline, team size, and complexity. The development team should also consider the customer's preferences and communication style when selecting an approach. Ultimately, the goal is to choose an approach that can provide the best results for the project while ensuring customer satisfaction.



[Fig: 4.8(a) Agile Software Development]

- Ultimately, the goal is to select tools and configurations that can provide optimal performance, reliability, and scalability for the system while also being cost-effective and easy to maintain.

5. System Design

5.1. System Design and Methodology

Designing a fuel management system web application requires a structured approach to ensure that it is efficient, reliable, and user-friendly. Here is a step-by-step methodology for designing a fuel management system web application.

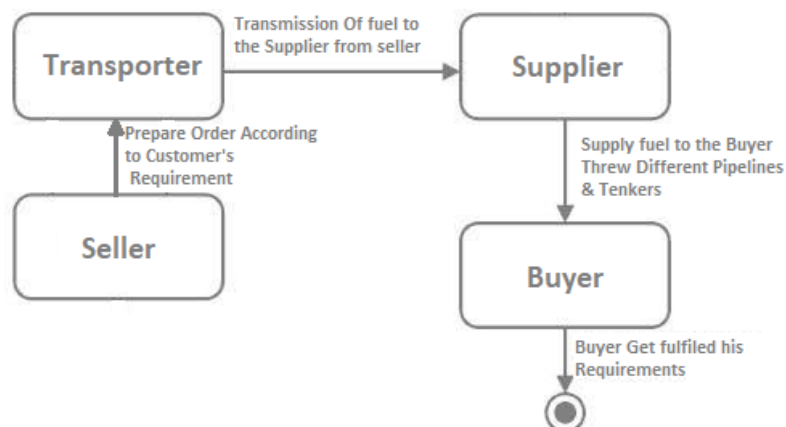
- **Requirements Gathering:** Gather and analyze the requirements of the fuel management system from the client or end-users. It is essential to understand the needs and expectations of the users and stakeholders to develop a system that satisfies their requirements.
- **System Architecture Design:** After gathering the requirements, create a system architecture design that outlines the structure and components of the system. This design should include the various modules, their interconnectivity, and the system's data flow.
- **Technology Stack Selection:** Select the technology stack to build the web application. This includes the programming languages, frameworks, and databases to be used. The technology stack should be chosen based on the system's requirements and its scalability, maintainability, and security.
- **UI/UX Design:** Design the user interface (UI) and user experience (UX) of the web application. The UI/UX design should be intuitive, easy to use, and visually appealing to the users.
- **Development:** Develop the web application using the selected technology stack, ensuring that the code is well-organized, easy to maintain, and scalable.
- **Testing:** Test the web application thoroughly to ensure that it meets the system's requirements, is user-friendly, and performs as expected.
- **Deployment:** Deploy the web application on a server, ensuring that it is secure, stable, and scalable. This involves setting up the server

infrastructure, configuring the web application, and ensuring that it is accessible to the users.

- **Maintenance and Support:** Provide ongoing maintenance and support for the web application to ensure that it remains secure, stable, and up-to-date with the latest technology. This includes fixing bugs, updating the software, and providing technical support to users.

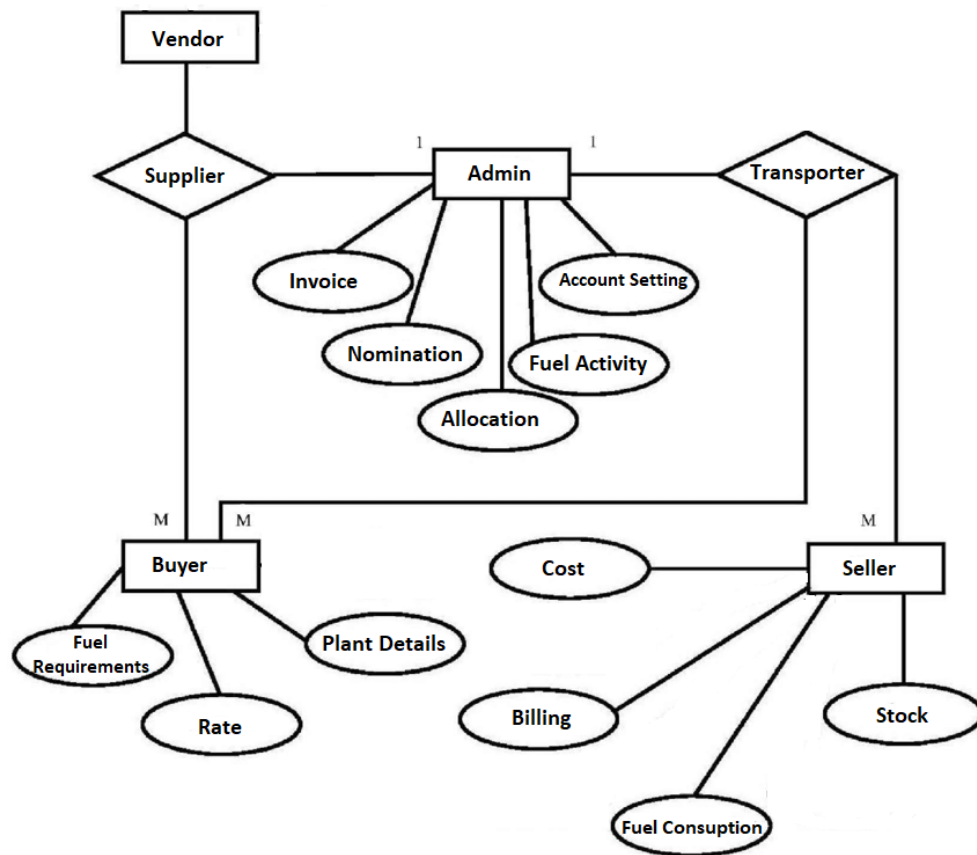
5.2. Input / Output and Interface Design

5.2.1. State Transition Diagram



[Fig: 5.2.1(a) State Transition Diagram]

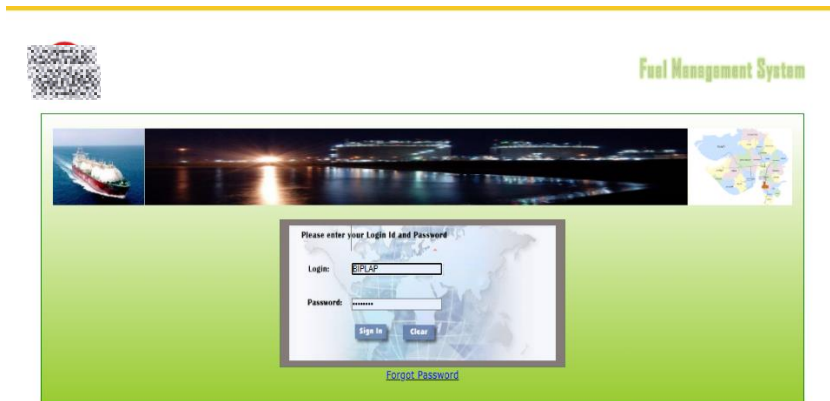
- ER Diagram :



[Fig: 5.2.1(b) ER Diagram]

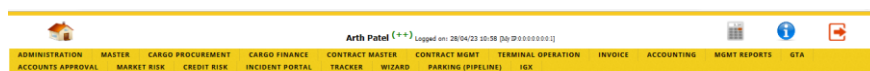
5.2.2. Sample of Interface :

Login Page :



[Fig: 5.2.2(a) Login Page]

Home Page :



[Fig: 5.2.2 (b) Home Page]

Add/Modify Group:

Arth Patel (++) Logged on: 28/04/23 10:58 [d/y P 00000000]

ADMINISTRATION MASTER CARGO PROCUREMENT CARGO FINANCE CONTRACT MASTER CONTRACT MGMT TERMINAL OPERATION INVOICE ACCOUNTING MGMT REPORTS GTA
ACCOUNTS APPROVAL MARKET RISK CREDIT RISK INCIDENT PORTAL TRACKER WIZARD PARKING (PIPELINE) IGX

ADMINISTRATION >> USER >> Add/Modify Group

ADD/MODIFY GROUP			ADD New Group
GROUP CODE	GROUP DESCRIPTION	ACTIVE / INACTIVE	
1	* ADMIN	☒	
2	* Comm Ops Coordinator	☒	
3	* FINANCE	☒	
4	* Read Only	☐	
5	* Supply Manager	☒	
6	* CFO & Dy CFO	☐	
7	* DATA Entry	☒	
8	* GM (MKTG. and COMM)	☐	
9	* FINANCE Tax	☒	
10	* FINANCE Treasury	☒	
11	* Marketing Manager	☐	
12	* Commercial Ops Manager	☒	
13	* Commercial (Exec)	☐	
14	* Sr. Marketing Advisor	☒	
15	* FINANCE Payment	☐	
16	* Risk Control Manager	☒	
17	* KYC	☒	
18	* Developer	☐	
19	* Credit Writing	☒	

Refresh * Mandatory Information Submit

[Fig: 5.2.2 (c) Add/Modify Group]

Add/Modify User :

ADD/MODIFY USER INFORMATION

User Name: User Id: E-mail Id:
 Group Name: From Date: 28/04/2023 To Date: 28/04/2024
 Company: Hazira Port Private Limited Remarks: Submit

USER DORMANT PROCESS

Inactive Tolerance Period: days.. Submit Recover Dormant Users

SELECT TO MODIFY FROM EMP LIST

USER NAME	LOGIN ID	EMAIL	GROUP	COMPANY	STATUS	LOCK STATUS	ACTIVITY
☐ Aarohi Dave	BIPLAD		KYC, ADMIN	SEIPL	Enabled	UnLocked	Reset Password
☐ Aarohi Dave	BIPLAS		Marketing Manager	SEIPL	Enabled	UnLocked	Reset Password
☐ Abhinav Trivedi	INATRF		Comm Ops Coordinator	SEIPL	Enabled	UnLocked	Reset Password
☐ Anupam Rungta	INARUI		DATA Entry	SEIPL	Enabled	UnLocked	Reset Password
☐ Arth Patel	BIPLAP		KYC, ADMIN, Risk Control Manager, Developer	HPPL	Enabled	UnLocked	Reset Password
☐ Arun Jain	INATF4		FINANCE	SEIPL	Enabled	UnLocked	Reset Password
☐ Ashish N. Shah	INASH5		ADMIN	SEIPL	Enabled	UnLocked	Reset Password
☐ Bhaumik Thakkar	INBTH2		FINANCE	SEIPL	Enabled	UnLocked	Reset Password
☐ Bhavik Shah	INBSH1		FINANCE Treasury	SEIPL	Enabled	UnLocked	Reset Password
☐ Chintan Joshi	INCJO0		Comm Ops Coordinator	SEIPL	Enabled	UnLocked	Reset Password
☐ Harsh Maheta	BIPLHM		KYC	HPPL	Enabled	UnLocked	Reset Password
☐ Jagrut Desai	INIDE2		FINANCE Treasury	SEIPL	Enabled	UnLocked	Reset Password
☐ Jagrut Shah	INJSH6		FINANCE Tax	SEIPL	Enabled	UnLocked	Reset Password
☐ Jayesh Patel	INJPA4		FINANCE	SEIPL	Enabled	UnLocked	Reset Password
☐ Krunal Pantagia	INKPBR		ADMIN	SEIPL	Enabled	UnLocked	Reset Password
☐ ...	...		Marketing Manager	SEIPL	Enabled	UnLocked	Reset Password

[Fig: 5.2.2 (d) Add/Modify User]

Group Allocation Delegation :

Arth Patel (++) Logged on: 28/04/23 10:58 [My ID:000000001]

GROUP ALLOCATION / DELEGATION

User Name:

Group Name:

From Date:

To Date:

ALLOCATED GROUPS TO ARTH PATEL

Select	Group	From Date	To Date
☐	KYC	23/03/2023	23/03/2024
☒	Developer	28/03/2023	28/03/2024
☐	ADMIN	29/03/2023	29/03/2024
☐	Risk Control Manager	29/03/2023	29/03/2024

[Fig: 5.2.2(e) Group Allocation Delegation]

User Details :

Arth Patel (++) Logged on: 28/04/23 10:58 [My ID:000000001]

ADMINISTRATION >> USER >> User Details

From: To:

☒ Enabled ☐ Disabled ☐ Dormant ☐ All

SR NO	USER NAME	LOGIN ID	DATE	EMAIL	STATUS	GROUP
1	Aarohi Dave	BIPLAD	23/03/2023	aarohi.dave@harvardinformatics.com	Enabled	KYC, ADMIN
2	Aarohi Dave	BIPLAS	21/04/2023		Enabled	Marketing Manager
3	Abhinav Trivedi	INATRF	02/07/2020		Enabled	Comm Ops Coordinator
4	Anupam Rungta	INARUI	27/02/2019		Enabled	DATA Entry
5	Arth Patel	BIPLAP	23/03/2023		Enabled	KYC, ADMIN, Risk Control Manager, Developer
6	Arun Jain	INAJF4	03/03/2020		Enabled	FINANCE
7	Ashish N. Shah	INASH5	26/03/2010		Enabled	ADMIN
8	Bhaumik Thakkar	INBTH2	14/08/2017		Enabled	FINANCE
9	Bhavik Shah	INBSH1	18/03/2016		Enabled	FINANCE Treasury
10	Chintan Joshi	INCIJ00	08/09/2011		Enabled	Comm Ops Coordinator
11	Harish Maheta	BIPLHM	19/04/2023		Enabled	KYC
12	Jagrut Desai	INJDE2	14/10/2010		Enabled	FINANCE Treasury
13	Jagrut Shah	INJSH6	25/03/2011		Enabled	FINANCE Tax
14	Jayesh Patel	INJPA4	17/08/2017		Enabled	FINANCE
15	Krunal Pastagia	INKPDR	30/01/2017		Enabled	ADMIN
16	Lalit Mohan	INLMO5	30/07/2019		Enabled	Marketing Manager
17	M Venkateshchalam	INMVE3	05/08/2010		Enabled	FINANCE
18	Medha Jaokar	INMJ06	18/04/2019		Enabled	FINANCE Treasury
19	Mehak Gupta	INMGV2	13/05/2020		Enabled	Commercial (Exec)
20	Milind Mehta	INMME1	05/08/2010		Enabled	FINANCE
21	Parinosh Shukla	INPSM4	25/01/2019		Enabled	Commercial Ops Manager
22	Pragati Gandhi	INPG06	07/05/2018		Enabled	FINANCE
23	Rajat Gupta	INRGUC	19/10/2010		Enabled	Marketing Manager
24	Renu Mam	BIPLRM	20/04/2023		Enabled	

[Fig: 5.2.2 (f) User Details]

Access Right Administration :

Arth Patel (++) Logged on: 28/04/23 10:58 [My ID: 0000000001]

ADMINISTRATION MASTER CARGO PROCUREMENT CARGO FINANCE CONTRACT MASTER CONTRACT MGMT TERMINAL OPERATION INVOICE ACCOUNTING MGMT REPORTS GTA

ACCOUNTS APPROVAL MARKET RISK CREDIT RISK INCIDENT PORTAL TRACKER WIZARD PARKING (PIPELINE) IGX

ADMINISTRATION >> USER >> Access Right Administration

Go Back Module Type : Form

ACCESS RIGHTS ADMINISTRATION

Group Name : ADMIN

MODULE NAME	FORM/REPORT NAME	SELECT	ACCESS RIGHTS LIST							
			READ	WRITE (PREPARED)	DELETE	PRINT	CHECK	AUTHORIZE	APPROVE	AUDIT
CONTRACT MGMT	Advance Payment Receipt	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Daily Allocation	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Daily Allocation(Buy Deal)	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Daily Buyer Nomination	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Daily Seller Nomination	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Daily Transporter Nomination	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Fortnightly Buyer Nomination	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Joint Ticket	Select	☒	☒	☒	☒	☒	☒	☒	☒
	L C Monitoring	Select	☒	☒	☒	☒	☒	☒	☒	☒
	L C Monitoring (Regas)	Select	☒	☒	☒	☒	☒	☒	☒	☒
	L C Monitoring (LTCORA & CN) □	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Liability Details	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Meter Ticket	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Monthly Buyer Nomination	Select	☒	☒	☒	☒	☒	☒	☒	☒
	Weekly Buyer Nomination	Select	☒	☒	☒	☒	☒	☒	☒	☒

Insert/Modify Access Rights

[Fig: 5.2.2 (g) Access Right Administration]

Cargo Bank :

Arth Patel (++) Logged on: 28/04/23 10:58 [My ID: 0000000001]

ADMINISTRATION MASTER CARGO PROCUREMENT CARGO FINANCE CONTRACT MASTER CONTRACT MGMT TERMINAL OPERATION INVOICE ACCOUNTING MGMT REPORTS GTA

ACCOUNTS APPROVAL MARKET RISK CREDIT RISK INCIDENT PORTAL TRACKER WIZARD PARKING (PIPELINE) IGX

CARGO PROCUREMENT >> CARGO MANAGEMENT >> Cargo Bank

CARGO BANK

REPORT DATE : 28/04/2023 DEAL TYPE : ALL

Cargo Price Modification >>> Cargo Allocation >>> Cargo Allocation Transfer >>> Expected Cargo Cancellation >>> Modify Pseudo Cargo Qty >>>

Report > Cargo Price Modification Reconcile Cargo Allocation >>> Report > Cargo Allocation Transfer Report > Cancelled Cargo Report > Pseudo Cargo Audit

UNLOADED CARGO	BALANCE QTY :	3567850.17	SHOW
OWN VOLUME ACCOUNT	BALANCE QTY :	321535.69	SHOW
EXPECTED CARGO	BALANCE QTY :	11351285.58	SHOW
PSEUDO CARGO	BALANCE QTY :	150000.00	SHOW

[Fig: 5.2.2 (h) Cargo Bank]

Supply Notice:

Arth Patel (++) Logged on: 28/04/23 10:58 [My IP:0000000001]

ADMINISTRATION MASTER CARGO PROCUREMENT CARGO FINANCE CONTRACT MASTER CONTRACT MGMT TERMINAL OPERATION INVOICE ACCOUNTING MGMT REPORTS GTA ACCOUNTS APPROVAL MARKET RISK

CREDIT RISK INCIDENT PORTAL TRACKER WIZARD PARKING (PIPELINE) IGX

CONTRACT MASTER >> Contract >> Supply Notice

SUPPLY NOTICE VERSION: 9.1.4 (TRANS ENTRY POINT: X-DTRANK)

Select Customer: BIPL Customer ID: 9 Contract ID: 9

SN NO: 2 View List

Revision: ☐ Apply Revision

* SN Name: BIPL-FG1-FGREV1-SN2-SHREV1

* Customer's Name: BIPL - BIPL LTD

* Start Date: 10/01/2022

* Sales Gas Price: 25.0000 (A.D.BTU) Request To Change Price Variable Pricing

* Quantity: TCQ: 70000 (A.D.BTU)

SN Closure: ☐ Effective Date: 31/12/2022 Request for SN Closure SN Closure Qty: (A.D.BTU)

Quantity Modification: TCQ: + (A.D.BTU) Request For TCQ Modification

SN Ref. NO:

Transportation Charge: (A.D.BTU)

Security: ☐ Incoming ☐ Security/Pre-Receipt

Variation: Total (A.D.BTU) For Whole Supply Period

Formula Remark:

Remarks: --Select--

* Delivery Point - SEIPL: ☐ SEIPL - Business Unit: ☐ SEIPL - Plant: ☒ GAIL ☐ Hazira-11n ☒ GAIL22 ☐ GAIL3 ☒ D6Gadimora ☐ Mhasakal ☐ GAIL5

* Plant(s): ☒ BIPL-Banda ☐ BIPL-Bombay

* Contract Base (Ex-Term / DLV based): ☐ Ex-Term ☒ Delivery Link Transporter Contracts

☐ Split TCQ: Firm QTY: (A.D.BTU)

* Deal Enter Date: 13/05/2022 13:10 Hrs:Min

* SIGNING DATE: 05/01/2022 00:00 HRS:MIN

Effective Date:

* End Date: 31/12/2022

* Allocated QTY: 19999

* DCQ: 7000 (A.D.BTU) Variable DCQ

Previous SN Ref. NO:

RE QTY: (A.D.BTU)

GOVERNING CLAUSES

☒ Buyer Nomination: ☐ Monthly ☐ Weekly ☒ Daily ☐ Fortnightly

☒ Seller Nomination: ☐ Monthly ☐ Weekly ☒ Daily ☐ Fortnightly

☐ Day Definition: From: 06:00 Hrs:Min To: 06:00 Hrs:Min

☒ MDCQ: 105 % of DCQ

Cutoff Time: 14:00 Hrs:Min

Activate Windows
Go to Settings to activate Windows.

[Fig: 5.2.2 (i) Supply Notice]

Seller's Nomination To Customer :

Arth Patel (++) Logged on: 28/04/23 10:58 [My IP:0000000001]

ADMINISTRATION MASTER CARGO PROCUREMENT CARGO FINANCE CONTRACT MASTER CONTRACT MGMT TERMINAL OPERATION INVOICE ACCOUNTING MGMT REPORTS GTA ACCOUNTS APPROVAL MARKET RISK

CREDIT RISK INCIDENT PORTAL TRACKER WIZARD PARKING (PIPELINE) IGX

CONTRACT MGMT >> Nomination >> Seller Nomination To Customer

SELLER'S DAILY NOMINATION TO CUSTOMER

GAS DAY: 16/04/2019 View

Print All Check All

CUSTOMER NAME	TO CONTACT PERSON	SN/LOA/CARGO NO	SEQ NO	PLANT NAME	QUANTITY (MMBTU)	FROM CONTACT PERSON	REMARKS	ACTIVITY-1	ACTIVITY-2	ACTIVITY-3	ACTIVITY-4
RIL	Reliance Industries Limited	CN (6) - 11	1	BIPL-Banda	850	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	1	BIPL-Banda	14000	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	1	BIPL-Banda	30000	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	1	BIPL-Banda	241000	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	2	BIPL-Banda	125	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	2	BIPL-Banda	450	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	1	BIPL-Banda	200000	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	2	BIPL-Banda	2800	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	2	BIPL-Banda	12000	Shaily Anand	Submit				
RIL	Reliance Industries Limited	CN (6) - 11	1	BIPL-Banda	5000	Shaily Anand	Submit				
GSFCL	Mr Devendra Aganwal	PERIOD (12018) - 1	1	BIPL-Banda	140000	Shaily Anand	Submit				
MNGL	Mr P. Pavan Kumar	SN - 6	1	BIPL-Banda	2500	Shaily Anand	Submit				

[Fig: 5.2.2(j) Sellers Nomination To Customer]

Arth Patel (++) Logged on: 26/04/23 10:58 My IP:0.0.0.0.0.0.1

ADMINISTRATION	MASTER	CARGO PROCUREMENT	CARGO FINANCE	CONTRACT MASTER	CONTRACT MGMT	TERMINAL OPERATION	INVOICE	ACCOUNTING	MGMT REPORTS	GTA	ACCOUNTS APPROVAL	MARKET RISK
CREDIT RISK	INCIDENT PORTAL	TRACKER	WIZARD	PARKING (PIPELINE)	IGX							

CONTRACT MGMT >> Nominations >> Fortnightly Buyer Nomination

FORTNIGHTLY BUYER NOMINATION (VERSION: 9.2.3)(TRANS ENTRY POINT)

GEN DAY: 26/04/2023 View

☐ FIRST FORTNIGHT ☒ SECOND FORTNIGHT

GCV: 9802.8 KCAL SCM NCV: 5831.35 KCAL SCM

Fortnight Start: 16/04/2023 and Fortnight End: 30/04/2023

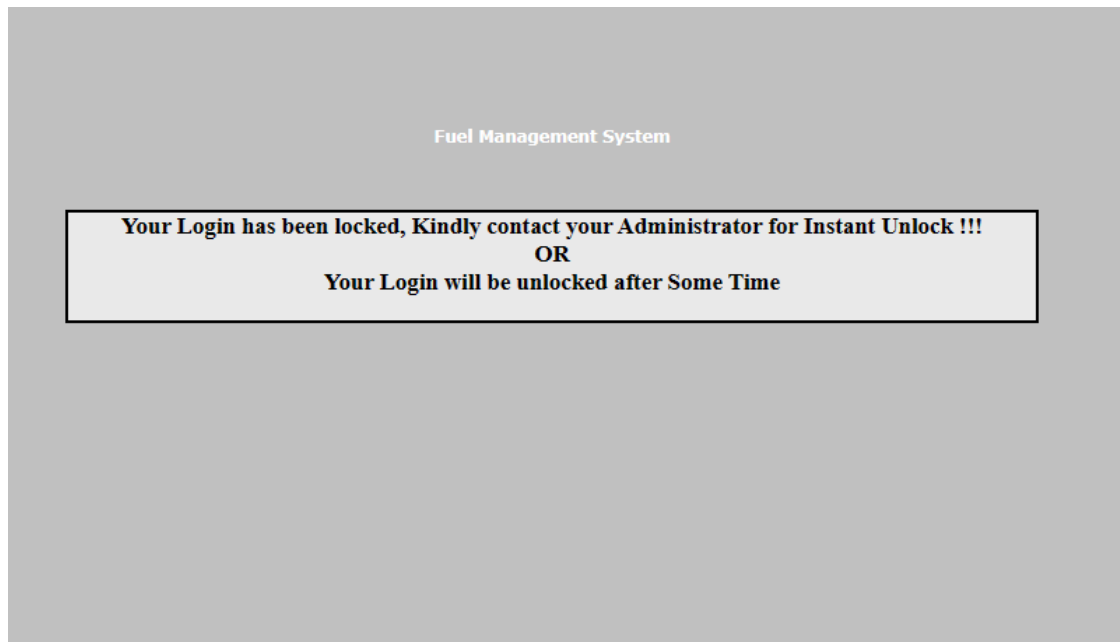
Time ITEM	SINCLA No. [RefNo.]	Plant Selection	DQO (M.B/T)	Total M.B/T	Total SCM	Generate Fortnightly Nomination
PIPELINE : GAIL :						
VGL : 13/04/2023 - 0						
VGL	16.00	SIN(S)	[Barcode]	140.00	1990	51156.79
		SIN(S)		140.00		
TOTAL (GAIL) :				280.00	1990.00	51156.79
GAIL3 :						
VGL		SIN(S)	[Barcode]	140.00		
VGL		SIN(S)		140.00		
TOTAL (GAIL3) :				280.00		
GAIL5 :						
VGL		SIN(S)	[Barcode]	140.00		
VGL		SIN(S)		140.00		
TOTAL (GAIL5) :				280.00		
SUM OF TOTAL :				840.00	1990.00	51156.79

Reset

FORTNIGHTLY BUYER NOMINATION DETAILS FOR GAIL			
CUSTOMER	FGSA NO.	SN NO.	GCV KCAL/SCM
VGL	1	5	9802.8
TIME	PLANT	DCQ (MMBTU)	NCV KCAL/SCM
16:00	VGL_Transpek Silox	140.00	8831.35
☐ DO YOU WANT TO REVISE DATA		SN START DATE : 10/04/2023	SN END DATE : 16/04/2023
DAY	DATE	MMBTU	SCM
SUNDAY	16/04/2023	130	3341.90
MONDAY	17/04/2023	139	3573.26
TUESDAY	18/04/2023	138	3547.56
WEDNESDAY	19/04/2023	120	3084.83
THURSDAY	20/04/2023	130	3341.90
FRIDAY	21/04/2023	138	3547.56
SATURDAY	22/04/2023	136	3496.14
SUNDAY	23/04/2023	135	3470.44
MONDAY	24/04/2023	133	3419.02
TUESDAY	25/04/2023	133	3419.02
WEDNESDAY	26/04/2023	133	3419.02
THURSDAY	27/04/2023	125	3213.37
FRIDAY	28/04/2023	130	3341.90
SATURDAY	29/04/2023	130	3341.90
SUNDAY	30/04/2023	140	3598.97
TOTAL		1990	409460.15

Close

Submit

User Lock Page :

[Fig: 5.2.2 (m) Lock User Page]

5.2.3. Database Tables :

- FMS7_SN_TRANSPORTER_DTL :

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	SN_NO	NUMBER(6,0)	No	(null)	1	(null)					
2	SN_REV_NO	NUMBER(2,0)	No	(null)	2	(null)					
3	FGSA_NO	NUMBER(6,0)	No	(null)	3	(null)					
4	FGSA_REV_NO	NUMBER(2,0)	No	(null)	4	(null)					
5	CUSTOMER_CD	NUMBER(6,0)	No	(null)	5	(null)					
6	TRANSPORTER_CD	NUMBER(6,0)	No	(null)	6	(null)					
7	EMP_CD	NUMBER(6,0)	Yes	(null)	7	(null)					
8	ENT_DT	DATE	Yes	(null)	8	(null)					
9	FLAG	CHAR(1 BYTE)	Yes	(null)	9	(null)					

[Fig: 5.2.3(a) FMS7_SN_TRANSPORTER_DTL]

- **HR_EMP_MST :**

Columns						
Data Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL						
▼ Actions...						
	1	2	3	4	5	6
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	EMP_CD	NUMBER(6,0)	No	(null)	1	(null)
2	EMP_ABR	VARCHAR2(15 BYTE)	Yes	(null)	2	(null)
3	JOIN_DT	DATE	Yes	(null)	3	(null)
4	EMP_NM	VARCHAR2(40 BYTE)	Yes	(null)	4	(null)
5	ADDR1	VARCHAR2(40 BYTE)	Yes	(null)	5	(null)
6	ADDR2	VARCHAR2(40 BYTE)	Yes	(null)	6	(null)
7	ADDR3	VARCHAR2(40 BYTE)	Yes	(null)	7	(null)
8	CITY	VARCHAR2(40 BYTE)	Yes	(null)	8	(null)
9	PIN	NUMBER(7,0)	Yes	(null)	9	(null)
10	QUAL_TN	VARCHAR2(40 BYTE)	Yes	(null)	10	(null)
11	SEX	CHAR(1 BYTE)	Yes	(null)	11	(null)
12	MARY_STATUS	CHAR(1 BYTE)	Yes	(null)	12	(null)
13	BIRTH_DT	DATE	Yes	(null)	13	(null)
14	BIRTH_PLACE	VARCHAR2(40 BYTE)	Yes	(null)	14	(null)
15	HEIGHT	VARCHAR2(6 BYTE)	Yes	(null)	15	(null)
16	WEIGHT	VARCHAR2(6 BYTE)	Yes	(null)	16	(null)
17	ES_LEFT	VARCHAR2(10 BYTE)	Yes	(null)	17	(null)
18	ES_RIGHT	VARCHAR2(10 BYTE)	Yes	(null)	18	(null)
19	RELIGION	VARCHAR2(20 BYTE)	Yes	(null)	19	(null)
20	DISABILITY_DTL	VARCHAR2(50 BYTE)	Yes	(null)	20	(null)
21	CRIME_DTL	VARCHAR2(50 BYTE)	Yes	(null)	21	(null)
22	INTEREST_DTL	VARCHAR2(50 BYTE)	Yes	(null)	22	(null)
23	HOME_STATE	VARCHAR2(30 BYTE)	Yes	(null)	23	(null)
24	PUNCHCARD_NO	VARCHAR2(4 BYTE)	Yes	(null)	24	(null)
25	PERM_ADDR1	VARCHAR2(40 BYTE)	Yes	(null)	25	(null)
26	PERM_ADDR2	VARCHAR2(40 BYTE)	Yes	(null)	26	(null)

[Fig: 5.2.3 (b) HR_EMP_MST]

- **SEC_EMP_GRP_DTL :**

Columns						
Data Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL						
▼ Actions...						
	1	2	3	4	5	6
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	EMP_CD	NUMBER(6,0)	Yes	(null)	1	(null)
2	GROUP_CD	NUMBER(3,0)	Yes	(null)	2	(null)
3	FROM_DT	DATE	Yes	(null)	3	(null)
4	TO_DT	DATE	Yes	(null)	4	(null)

[Fig: 5.2.3 (c) SEC_EMP_GRP_DTL]

- **FMS7_TRANSPORTER_MST :**

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
▼ Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	TRANSPORTER_CD	NUMBER(6,0)	No	(null)	1 (null)						
2	EFF_DT	DATE	No	(null)	2 (null)						
3	TRANSPORTER_...	VARCHAR2(100 BYTE)	Yes	(null)	3 (null)						
4	TRANSPORTER_...	VARCHAR2(10 BYTE)	Yes	(null)	4 (null)						
5	GST_TIN_NO	VARCHAR2(20 BYTE)	Yes	(null)	5 (null)						
6	GST_TIN_DT	DATE	Yes	(null)	6 (null)						
7	CST_TIN_NO	VARCHAR2(20 BYTE)	Yes	(null)	7 (null)						
8	CST_TIN_DT	DATE	Yes	(null)	8 (null)						
9	PAN_NO	VARCHAR2(20 BYTE)	Yes	(null)	9 (null)						
10	PAN_ISSUE_DT	DATE	Yes	(null)	10 (null)						
11	TAN_NO	VARCHAR2(20 BYTE)	Yes	(null)	11 (null)						
12	TAN_ISSUE_DT	DATE	Yes	(null)	12 (null)						
13	ADDL_NO	VARCHAR2(20 BYTE)	Yes	(null)	13 (null)						
14	ADDL_ISSUE_DT	DATE	Yes	(null)	14 (null)						
15	WEB_ADDR	VARCHAR2(100 BYTE)	Yes	(null)	15 (null)						
16	NOTES	VARCHAR2(2048 B...	Yes	(null)	16 (null)						
17	EMP_CD	NUMBER(6,0)	Yes	(null)	17 (null)						
18	ENT_DT	DATE	Yes	(null)	18 (null)						
19	FLAG	CHAR(1 BYTE)	Yes	(null)	19 (null)						
20	GSTIN_NO	VARCHAR2(15 BYTE)	Yes	(null)	20 (null)						
21	GSTIN_DT	DATE	Yes	(null)	21 (null)						
22	DORMANT_FLAG	CHAR(1 BYTE)	Yes	(null)	22 (null)						
23	COUNTERPARTY_CD	NUMBER(6,0)	Yes	(null)	23 (null)						

[Fig: 5.2.3 (d) FMS7_TRANSPORTER_MST]

- **SEC_ACCESS_GROUP_MST :**

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
▼ Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	GROUP_CD	NUMBER(3,0)	No	(null)	1 (null)						
2	GROUP_NAME	VARCHAR2(50 BYTE)	Yes	(null)	2 (null)						
3	ACTIVE	CHAR(1 BYTE)	Yes	(null)	3 (null)						
4	EFF_DATE	DATE	Yes	(null)	4 (null)						

[Fig: 5.2.3 (e) SEC_ACCESS_GROUP_MST]

- **FMS7_FN_BUYER_NOM_DTL :**

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
▼ Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	SN_NO	NUMBER (6,0)	No	(null)	1	(null)					
2	FGSA_NO	NUMBER (6,0)	No	(null)	2	(null)					
3	NOM_REV_NO	NUMBER (2,0)	No	(null)	3	(null)					
4	TRANSPORTER_CD	NUMBER (6,0)	No	(null)	4	(null)					
5	CUSTOMER_CD	NUMBER (6,0)	No	(null)	5	(null)					
6	PLANT_SEQ_NO	NUMBER (2,0)	No	(null)	6	(null)					
7	GAS_DT	DATE	No	(null)	7	(null)					
8	CONTRACT_TYPE	CHAR (1 BYTE)	No	(null)	8	(null)					
9	GEN_DT	DATE	No	(null)	9	(null)					
10	GEN_TIME	VARCHAR2 (6 BYTE)	No	(null)	10	(null)					
11	GCV	NUMBER (7,2)	No	(null)	11	(null)					
12	NCV	NUMBER (7,2)	No	(null)	12	(null)					
13	QTY_MMBTU	NUMBER (9,2)	No	(null)	13	(null)					
14	QTY_SCM	NUMBER (11,2)	No	(null)	14	(null)					
15	SN_DOC_NO	VARCHAR2 (10 BYTE)	Yes	(null)	15	(null)					
16	EMP_CD	NUMBER (6,0)	Yes	(null)	16	(null)					
17	ENT_DT	DATE	Yes	(null)	17	(null)					
18	FLAG	CHAR (1 BYTE)	Yes	(null)	18	(null)					
19	MAPPING_ID	VARCHAR2 (40 BYTE)	Yes	(null)	19	(null)					

[Fig: 5.2.3 (f) FMS7_FN_BUYER_NOM_DTL]

- **SEC_FORM_MST :**

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
▼ Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	FORM_CD	NUMBER (4,0)	No	(null)	1	(null)					
2	FORM_NAME	VARCHAR2 (50 BYTE)	Yes	(null)	2	(null)					
3	MODULE_CD	NUMBER (2,0)	Yes	(null)	3	(null)					
4	CLASSPATH	VARCHAR2 (60 BYTE)	Yes	(null)	4	(null)					
5	DEPLOY_DT	DATE	Yes	(null)	5	(null)					
6	DOC_TYPE	VARCHAR2 (1 BYTE)	Yes	(null)	6	(null)					
7	FLAG	VARCHAR2 (1 BYTE)	Yes	(null)	7	(null)					
8	GRP_CD	NUMBER (4,0)	Yes	(null)	8	(null)					
9	GRP_NM	VARCHAR2 (20 BYTE)	Yes	(null)	9	(null)					
10	GRP_SEQ_NO	NUMBER (2,0)	Yes	(null)	10	(null)					

[Fig: 5.2.3 (g) SEC_FORM_MST]

- **SEC_EMP_PASSWORDS :**

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	EMP_CD	NUMBER(6,0)	No	(null)	1	(null)					
2	PASSWORD	VARCHAR2(50 BYTE)	Yes	(null)	2	(null)					
3	LOGIN_STATUS	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)					
4	PASSWORD1	VARCHAR2(50 BYTE)	Yes	(null)	4	(null)					
5	PASSWORD2	VARCHAR2(50 BYTE)	Yes	(null)	5	(null)					
6	LAST_CHANGE	DATE	Yes	(null)	6	(null)					
7	LOCKED_FLAG	VARCHAR2(1 BYTE)	Yes	(null)	7	(null)					
8	PASSWORD3	VARCHAR2(50 BYTE)	Yes	(null)	8	(null)					
9	RESET_FLAG	VARCHAR2(1 BYTE)	Yes	(null)	9	(null)					
10	PASSWORD4	VARCHAR2(50 BYTE)	Yes	(null)	10	(null)					
11	PASSWORD5	VARCHAR2(50 BYTE)	Yes	(null)	11	(null)					
12	PASSWORD6	VARCHAR2(50 BYTE)	Yes	(null)	12	(null)					
13	PASSWORD7	VARCHAR2(50 BYTE)	Yes	(null)	13	(null)					
14	PASSWORD8	VARCHAR2(50 BYTE)	Yes	(null)	14	(null)					
15	PASSWORD9	VARCHAR2(50 BYTE)	Yes	(null)	15	(null)					
16	PASSWORD10	VARCHAR2(50 BYTE)	Yes	(null)	16	(null)					
17	EXPIRY_DATE	DATE	Yes	(null)	17	(null)					
18	LOCKED_DATE	DATE	Yes	(null)	18	(null)					

[Fig: 5.2.3 (h) SEC_EMP_PASSWORDS]

- **FMS9_COUNTERPARTY_MST :**

Columns	Data	Constraints	Grants	Statistics	Triggers	Flashback	Dependencies	Details	Partitions	Indexes	SQL
Actions...											
	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS					
1	COUNTERPARTY_CD	NUMBER(6,0)	No	(null)	1	(null)					
2	EFF_DT	DATE	No	(null)	2	(null)					
3	COUNTERPARTY_NM	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)					
4	COUNTERPARTY...	VARCHAR2(10 BYTE)	Yes	(null)	4	(null)					
5	PAN_NO	VARCHAR2(20 BYTE)	Yes	(null)	5	(null)					
6	PAN_ISSUE_DT	DATE	Yes	(null)	6	(null)					
7	KYC_NOTES	VARCHAR2(2048 B...	Yes	(null)	7	(null)					
8	ENT_BY	NUMBER(6,0)	Yes	(null)	8	(null)					
9	ENT_DT	DATE	Yes	(null)	9	(null)					
10	MODIFY_BY	NUMBER(6,0)	Yes	(null)	10	(null)					
11	MODIFY_DT	DATE	Yes	(null)	11	(null)					
12	FLAG	CHAR(1 BYTE)	Yes	(null)	12	(null)					
13	DORMANT_FLAG	CHAR(1 BYTE)	Yes	(null)	13	(null)					
14	DORMANT_CHEC...	CHAR(1 BYTE)	Yes	(null)	14	(null)					
15	KYC	CHAR(1 BYTE)	Yes	(null)	15	(null)					
16	IGX	CHAR(1 BYTE)	Yes	(null)	16	(null)					

[Fig: 5.2.3 (i) FMS9_COUNTERPARTY_MST]

6. Implementation

6.1.Implementation Of Platform :

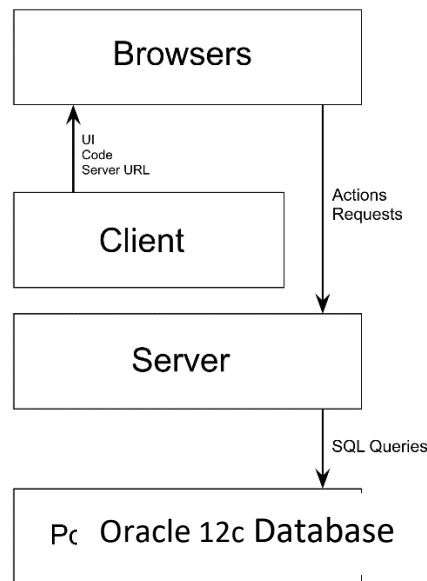
Eclipse IDE:

- We need eclipse IDE as a Integrated Development Environment. The Eclipse IDE is famous for our Java Integrated Development Environment (IDE), but we have a number of pretty cool IDEs, including our C/C++ IDE, JavaScript/TypeScript IDE, PHP IDE, and more.
- You can easily combine multiple languages support and other features into any of our default packages, and the Eclipse Marketplace allows for virtually unlimited customization and extension.
- Develop your software wherever you go. It'll be there, in the cloud, right where you left it. Use your browser to develop with hosted workspaces or install desktop packaging to experience a modern development environment for Java, JavaScript, CSS, and HTML.

Oracle 12c (Database) :

- Oracle released Oracle Database 12c into general availability July 1, 2013. According to Oracle, this is "the first database designed for the cloud." Oracle Database 12c also introduces 500 new features to the database, most notably pluggable databases and multitenant architecture.
- Oracle Database 12c is a relational database management system (RDBMS) designed for both on-premises and cloud uses and deployable on a choice of clustered or single servers. It provides comprehensive features for managing data in transaction processing, business intelligence and content management applications.
- An added benefit of performing the migration in Oracle Database 12c is that the new extended VARCHAR2 type limit of 32767 bytes guarantees any VARCHAR2 column with the pre-12.1 limit of 4000 bytes or less can just be lengthened to accommodate longer values resulting from data expanding in conversion to Unicode.

6.2.Process / Technology / Module Specifications



[Fig: 6.2 (a) cross platform]

6.3. Results / Outcomes

Finally, as outcomes we got a web app file. Which can be run in any Browser. All the files of project will be run on any browser . Outcome of the project will be shown as per the requirement.

6.4. Result Analysis

The SRS & BRS was provided by BIPL. As we processed the project, we matched the SRS provide by company . During the review conducted by mentor thechanges which was suggested by the mentor were implemented by us in the project.All modules and Features are listed present in the final application.

Application is running on large scale

7. Testing

As your codebase expands, small errors and edge cases you don't expect can cascade into larger failures. Bugs lead to bad user experience and ultimately, business losses. One way to prevent fragile programming is to test your code before releasing it into the wild.

We're humans, and humans make mistakes. Testing is important because it helps you uncover these mistakes and verifies that your code is working. Perhaps even more importantly, testing ensures that your code continues to work in the future as you add new features, refactor the existing ones, or upgrade major dependencies of your project. There is more value in testing than you might realize. One of the best ways to fix a bug in your code is to write a failing test that exposes it. Then when you fix the bug and re-run the test, if it passes it means the bug is fixed, never reintroduced into the code Base.

Tests can also serve as documentation for new people joining your team. For people who have never seen a codebase before, reading tests can help them understand how the existing code works.

7.1. Testing Plan / Strategy

7.1.1. Alpha Testing

The Phase "Alpha Testing" is carried out by the developer himself. Along With the programming developer have to test his work. This is very important for the developer to solve the errors along with the programming So developer can develop a better solution & error free software.

7.1.2. Beta Testing

The Phase "Beta Testing" is done by the senior developer or the project manager. In this type of testing manager or tester Checks all the possible outcomes & ways of program execution. So that some of the errors can be

found on the development stage & can be solved.

7.1.3. FAT Testing

The Phase “FAT Testing” is carried Out by the Test team of the Company . In this phase , test team will be check all the possibilities & try to find as much as errors so they can be resolved before send the software to the user.

7.1.4. UAT Testing

The Phase “UAT Testing” is carried out by the user . After FAT Testing software will be given to the User. In which user Will Test the Software according to his Requirement & He will check all the possibilities of outcome & result.

7.2. Test Result and Analysis

After Writing Alpha Testing , Beta Testing , FAT Testing & UAT testing

7.2.1. Test Cases

- All functions and there return types are tested with Alpha testing
- Beta testing and their response testing
- All Screens render tests
- Data Integrity tests

8. Conclusion and Discussion

8.1. Conclusion / Overall Analysis

My internship provided me with a wealth of valuable experiences, including opportunities to learn new knowledge and skills, connect with new people, and gain insight into professional practice. While I was able to achieve several of my learning goals, some were not feasible given the conditions. During my time at the company, I gained a comprehensive understanding of the different facets of working within an organization, including the importance of financing in driving project progress. Additionally, I deepened my knowledge of project development in Java, although I recognize that there is still much to learn and improve upon, as the methods used are not yet standardized. Through this experience, I also discovered my strengths and weaknesses, which will allow me to better focus my efforts in improving my skills and knowledge in the future. Overall, I believe that this internship was a valuable experience that will have a positive impact on my professional growth.

8.2. Photographs and date of surprise visit by institute mentor

- There is No visit by the institute.

8.3. Dates of Continuous Evaluation

There were no evaluations carried out.

8.4. Problem Encounter and Possible Solution

During my internship, I encountered various challenges that were new to me, which sometimes made things difficult. However, I was fortunate to have a supportive team, including my mentor and colleagues, who were always willing to help me out. Whenever I faced fresh challenges, my seniors and mentors were there to offer advice and guidance, which allowed me to overcome these obstacles. Thanks to their support and expertise, I was able to learn from my experiences and continue to grow both personally and professionally.

8.5. Summary of Internship Project

The Fuel Management System is a software system that provides an efficient way to manage the supply and storage of fuel for different entities such as customers, transporters, suppliers, and vendors. The system is designed to make it easier for users to initiate their fuel requirements by entering into buy/sell contracts with energy suppliers.

The system operates through different modules that contain forms and reports which provide a user-friendly interface that is consistent throughout the application. The validation checks are also included to enable end-users to avoid operational mistakes. To operate the Fuel Management System, users first create a customer entity within the system. After the customer entity is created, the customer and supplier sign a deal for fuel supply as per the customer's requirement.

The technologies used to develop the Fuel Management System include Apache Tomcat 8.0, J2EE, and Oracle 12c database. Apache Tomcat is an open-source web server and servlet container used to serve Java web pages, while J2EE (Java 2 Platform, Enterprise Edition) is a programming platform for developing and deploying enterprise applications.

The Oracle 12c database is used to store and manage the data within the Fuel Management System. It is a relational database management system that provides a wide range of features for efficient data management.

Finally, the project assignment for the Fuel Management System involves enhancements and technical support. This means that the system will continue

to be improved and updated based on feedback and changing needs, and users will have access to technical support to help them resolve any issues they encounter while using the system.

8.6. Limitation and Future Enhancements

The fuel management system is highly susceptible to market fluctuations and changes in oil and gas rates, which makes it challenging to track and manage this data in real-time. Additionally, there may be variations in the prices and rates of oil and gas across different regions, and the range of variation for LNG can be quite vast. Furthermore, there are limitations to the system, as significant changes in fuel management or buying/selling prices can occur at any time. Therefore, it is essential to monitor the system closely and implement enhancements as needed to ensure that it remains up-to-date and effective in managing fuel resources.

There are several possible enhancements that could be made to the fuel management system software to help address the challenges mentioned. Here are a few examples:

- **Real-time data integration:**

The system could be enhanced to integrate real-time data from various sources, such as oil and gas price trackers, weather forecasts, and traffic patterns. This would allow the system to better predict fuel needs and adjust inventory levels accordingly.

- **Regional pricing adjustments:**

The software could be designed to automatically adjust pricing based on regional variations in oil and gas prices. This would ensure that the system stays competitive and profitable across all markets.

- **Machine learning algorithms:**

The system could be enhanced with machine learning algorithms to help identify patterns and trends in fuel usage and market data. This would enable the system to make more accurate predictions and optimize fuel management strategies.

- **Automated alerts:**

The software could be enhanced with automated alerts that notify fuel managers of significant changes in fuel management or pricing. This would allow managers to take action quickly and efficiently.

9. References

- W3Schools (<https://www.w3schools.com/>): A popular online resource for learning web development. It covers a wide range of topics, including HTML, CSS, JavaScript, PHP, SQL, and more.
- Codecademy (<https://www.codecademy.com/>): Offers interactive online courses on a variety of programming languages and topics, including HTML, CSS, JavaScript, Java, and more.
- Udemy (<https://www.udemy.com/>): Offers a wide range of online courses, including many on web development, Java, HTML, and JavaScript.
- Coursera (<https://www.coursera.org/>): Offers online courses and specializations from top universities and organizations, including courses on web development, Java programming, and more.
- Mozilla Developer Network (<https://developer.mozilla.org/en-US/>): A comprehensive online resource for web developers, with documentation and guides on HTML, CSS, JavaScript, and more.
- Oracle Java Tutorials (<https://docs.oracle.com/en/java/javase/index.html>): A series of free online tutorials offered by Oracle, covering various aspects of Java programming.
- JavaScript.info (<https://javascript.info/>): A comprehensive resource for learning JavaScript, with interactive examples and exercises.
- Stack Overflow (<https://stackoverflow.com/>): An online community where developers can ask and answer programming questions, including those related to web development, Java, HTML, and JavaScript.